CHAPTER 4

Limits of functions

S21, S23, S24, S26

> **Key point · Objectives.** Calculate limits at a point and at infinity; resolve an indeterminate form; interpret an asymptote.

Course notes

Meaning of a limit

$f(x) \to \ell$ as $x \to +\infty$ means that $f(x)$ can be made as close as desired to ℓ for all sufficiently large x. For a limit at a, we consider values of x in the domain, close to a and different from a. The left-hand and right-hand limits are written $\lim_{x\to a^-} f(x)$ and $\lim_{x\to a^+} f(x)$. When the domain has points on both sides of a, a two-sided limit exists if and only if both one-sided limits exist and are equal. At an endpoint of the domain, we study the limit from the only accessible side.

> **Key point · Graphical interpretation.** If $f(x) \to \ell \in \mathbb{R}$ as $x \to +\infty$ or $-\infty$, the line $y = \ell$ is a horizontal asymptote in that direction. If $f(x) \to \pm\infty$ as $x \to a$ from one side, the line $x = a$ is a vertical asymptote.

A curve can cross a horizontal asymptote. The limit does not necessarily give the position of the curve relative to the asymptote: study the sign of $f(x) - \ell$.

Reference limits and operations

At $+\infty$, $x^m \to +\infty$ for $m \geq 1$, $\frac{1}{x^m} \to 0$, $\sqrt{x} \to +\infty$. At $-\infty$, the sign of x^m depends on the parity of m. The rules for sum, product, quotient and comparison are those for sequences, with the same indeterminate forms.

Example. For $f(x) = \frac{2x+1}{x-3}$, write $f(x) = 2 + \frac{7}{x-3}$. The limits at $\pm\infty$ equal 2. At 3^-, $f \to -\infty$; at 3^+, $f \to +\infty$. The asymptotes are $x = 3$ and $y = 2$.

Conjugation. For $x > 0$, $\sqrt{x^2+4x} - x = \frac{4}{\sqrt{1+\frac{4}{x}}+1} \to 2$. For $x < 0$, take care: $\sqrt{x^2} = |x| = -x$.

Exponential and comparison of growth rates

$$\lim_{x\to+\infty} e^x = +\infty, \quad \lim_{x\to-\infty} e^x = 0, \quad \lim_{x\to+\infty} \frac{e^x}{x^m} = +\infty \quad (m \in \mathbb{N}).$$

We deduce $x^m e^{-x} \to 0$ at $+\infty$.

Proof of the limits of the exponential. Studying $e^x - x - 1$ gives $e^x \geq 1 + x$. By comparison, $e^x \to +\infty$ at $+\infty$. Setting $t = -x \to +\infty$, $e^x = \frac{1}{e^t} \to 0$.

Proof of a comparison of growth rates. For every integer $k \geq 0$ and $x \geq 0$, $e^x \geq \frac{x^k}{k!}$. For $k = 0$, $e^x \geq 1$. If the inequality holds at order k, the derivative of $e^x - \frac{x^{k+1}}{(k+1)!}$ is $e^x - \frac{x^k}{k!} \geq 0$; its value at 0 is positive. The inductive step follows. Choosing $k = m + 1$, $\frac{e^x}{x^m} \geq \frac{x}{(m+1)!} \to +\infty$.

> **Method · Choosing a transformation.** Quotient of polynomials: factorise the highest powers. Difference of roots: conjugate. Quotient near a zero: factorise and study the signs. Exponential against power: comparison of growth rates, after a change of variable if necessary.

Comparison and composition: conditions of use

If $g(x) \leq f(x) \leq h(x)$ near the point or the infinity under study, and if g, h have the same real limit ℓ, then $f(x) \to \ell$ (squeeze theorem). If $g \leq f$ and $g \to +\infty$, then $f \to +\infty$. For a limit of $-\infty$, use an upper bound by a function tending to $-\infty$. If $u(x) \to a$ and v is continuous at a, then $v(u(x)) \to v(a)$, whenever the composite is defined. For example $\sqrt{1 + \frac{1}{x}} \to 1$ at $+\infty$. For an infinite limit of the outer function, check from which side $u(x)$ approaches the bound: $\frac{1}{u(x)}$ does not behave in the same way depending on the sign of $u(x)$.

Research activity

A curve close to a line — 30 min.

Consider the function f defined on $\mathbb{R} \setminus \{1\}$ by $f(x) = \frac{3x-2}{x-1}$. Let $\mathcal{C}_f$ be its curve in a plane frame of reference.

1. Show that, for every $x \neq 1$, $f(x) = 3 + \frac{1}{x-1}$.
2. Determine the limits of f at $-\infty$, at 1^-, at 1^+ and at $+\infty$.
3. Study the sign of $f(x) - 3$, then state the position of $\mathcal{C}_f$ relative to the line with equation $y = 3$.
4. Determine a real number $A > 1$ such that, for every $x > A$, $|f(x) - 3| < 0.01$.
5. Does the curve $\mathcal{C}_f$ cross its horizontal asymptote? Justify, and explain why a graphing window that is too wide can give a misleading impression.

Progressive exercises

E1 · Practice. Determine $\lim\limits_{x\to+\infty} \dfrac{5x^3 - x}{2x^3 + 7}$, with justification.

E2 · Practice. Let f be the function defined on $\mathbb{R} \setminus \{2\}$ by $f(x) = \frac{1}{(x-2)^2}$. Determine its limits at 2^-, at 2^+, at $-\infty$ and at $+\infty$.

E3 · Practice. Determine $\lim\limits_{x\to1} \dfrac{x^2 - 1}{x - 1}$, stating why the simplification used is valid in a neighbourhood of 1.

E4 · Practice. Determine $\lim\limits_{x\to+\infty} x^3 e^{-2x}$. State the result on comparison of growth rates that is used.

E5 · Practice. Determine $\lim\limits_{x\to-\infty} \left(\sqrt{x^2+1} + x\right)$. Justify in particular the treatment of $\sqrt{x^2}$.

Textbook or past-paper work

Barbazo, p. 96, exercises 26 and 29. In S23: exercise 29, to link limits and asymptotes. At home after S24: exercise 26, to construct three counterexamples to an invalid rule for calculating with limits.

Homework

Homework 3 — A family of curves — 45 min.

For each real number a, consider the function f_a defined by $f_a(x) = \frac{ax^2+1}{x^2+1}$. Let $\mathcal{C}_a$ be its graph.

1. Determine the domain of f_a, then its limits at $-\infty$ and at $+\infty$.
2. Show that $f_a(x) - a = \frac{1-a}{x^2+1}$. Depending on the value of a, state the position of $\mathcal{C}_a$ relative to the line with equation $y = a$.
3. Study the variations of f_a by distinguishing the cases $a < 1$, $a = 1$ and $a > 1$. State any extrema and where they are attained.
4. Take $a = 3$. Solve, for $x \geq 0$, the inequality $|f_3(x) - 3| \leq 0.02$.